

```
In [1]: import pandas as pd
import numpy as np
import seaborn as sns
import matplotlib.pyplot as plt
```

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In [3]: from sklearn.model_selection import train_test_split
from sklearn.linear_model import LogisticRegression
from sklearn.preprocessing import StandardScaler
from sklearn.metrics import confusion_matrix, accuracy_score, precision_score, recall_score
```

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In [4]: df = pd.read_csv('Social_Network_Ads.csv')
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In [5]: df
```

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Out[5]:
```

	User ID	Gender	Age	EstimatedSalary	Purchased
0	15624510	Male	19	19000	0
1	15810944	Male	35	20000	0
2	15668575	Female	26	43000	0
3	15603246	Female	27	57000	0
4	15804002	Male	19	76000	0
...	...	...	...	...	...
395	15691863	Female	46	41000	1
396	15706071	Male	51	23000	1
397	15654296	Female	50	20000	1
398	15755018	Male	36	33000	0
399	15594041	Female	49	36000	1

400 rows × 5 columns

```
In [6]: df.columns
```

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Out[6]: Index(['User ID', 'Gender', 'Age', 'EstimatedSalary', 'Purchased'], dtype='object')
```

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In [7]: df.isnull().sum()
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```
Out[7]: User ID      0
Gender      0
Age         0
EstimatedSalary  0
Purchased   0
dtype: int64
```

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In [8]: df['Gender'] = df['Gender'].map({'Male':0,'Female':1})
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```
In [11]: X = df[['Gender','Age','EstimatedSalary']]
y = df['Purchased']

x_train,x_test,y_train,y_test = train_test_split(X,y,test_size = 0.25,random_state=
```

```
In [13]: sc = StandardScaler()
x_train = sc.fit_transform(x_train)
x_test = sc.transform(x_test)
```

```
In [14]: clfr = LogisticRegression()
clfr.fit(x_train,y_train)
y_pred = clfr.predict(x_test)
```

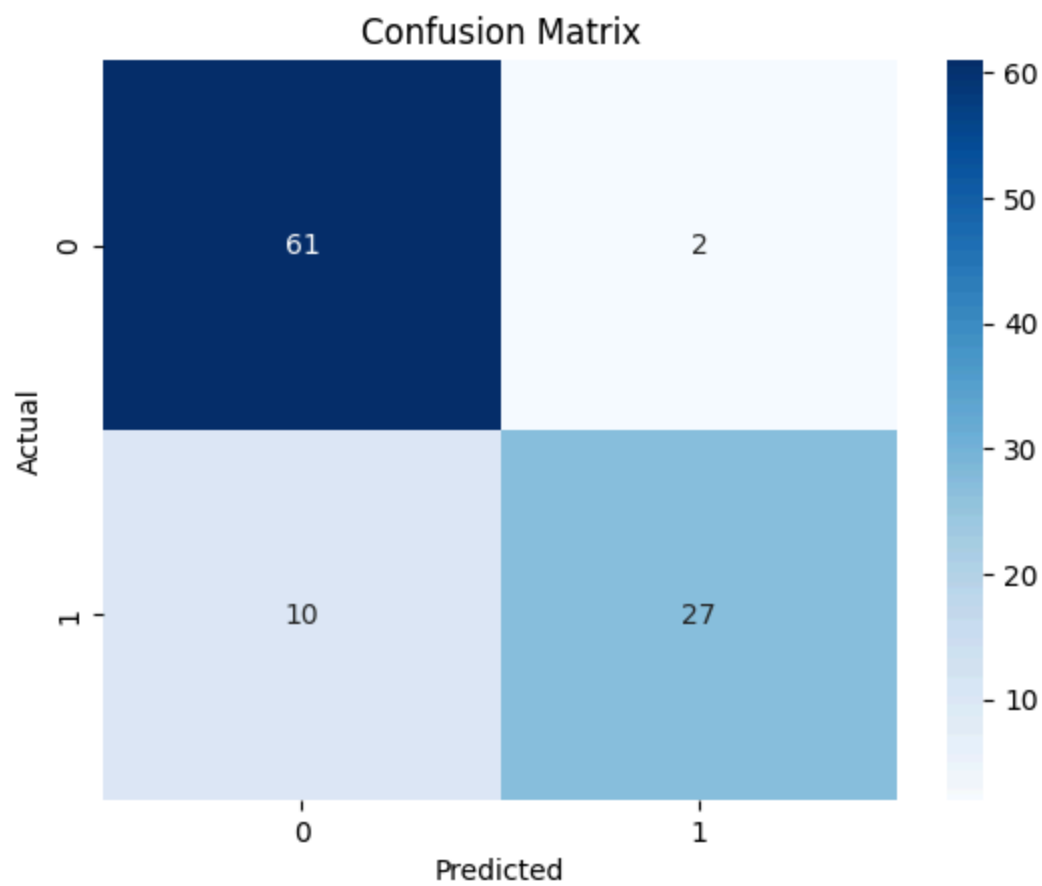
```
In [16]: cm = confusion_matrix(y_test,y_pred)
print("Confusion Matrix : ", cm)

TN, FP,FN,TP = cm.ravel()
accuracy = accuracy_score(y_test,y_pred)
error_rate = 1 - accuracy
precision = precision_score(y_test,y_pred)
recall = recall_score(y_test,y_pred)

print("True negative: ",TN)
print("False Positive: ",FP)
print("False negative: ",FN)
print("True positive: ",TP)
print("Accuracy score: ",accuracy)
print("Error rate: ",error_rate)
print("Precision score: ",precision)
print("Recall score: ",precision)
```

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Confusion Matrix : [[61  2]
 [10 27]]
True negative: 61
False Positive: 2
False negative: 10
True positive: 27
Accuracy score: 0.88
Error rate: 0.12
Precision score: 0.9310344827586207
Recall score: 0.9310344827586207
```

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In [22]: sns.heatmap(cm,annot=True,fmt="d",cmap="Blues")
plt.title('Confusion Matrix')
plt.xlabel("Predicted")
plt.ylabel("Actual")
plt.show()
```



In []: